EthMuSAM : Mutual Scoring With SAM For Zero Shot Anomaly Segmentation

Industrial Track Zero-shot Submission

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Abstract

In this paper, we present an enhanced hybrid framework for zero-shot visual anomaly detection, focusing on robust defect segmentation in complex real-world industrial environments. To significantly improve pixel-level anomaly localization, we integrate a cascaded prompt refinement strategy leveraging the Segment Anything Model (SAM) with the MuSc-DINOv3 architecture. Specifically, we extract initial anomaly heatmaps and employ a 3-pass prompting mechanism. By dynamically sampling positive and negative point prompts through adaptive Otsu thresholding and structural dilation, and generating refined bounding box prompts from connected components, our method effectively isolates defective regions. This integration seamlessly converts coarse anomaly heatmaps into precise binary segmentation masks using a parameter-free SAM refinement module. Experimental results on MVTec AD 2 demonstrate strong localization behavior, achieving a mean SegF1 of 27.78% *and several strong category-level results, including* 55.55% *SegF1 on fabric,* 80.94% *pixel-level AUROC on walnuts,* 94.67% *image-level AUROC on vial, and* 73.45% *mean image-level AP across categories.*

1. Introduction

1.1. Background

Industrial anomaly detection plays a critical role in maintaining operational efficiency, safety and rigorous quality control in modern manufacturing. By identifying defects early, automated visual inspection systems prevent costly system failures and production bottlenecks. Traditional supervised and even modern unsupervised (one-class) learning methods require collecting and training on extensive defect free samples for each new product category. However, in dynamic industrial environments with constantly shifting product lines, acquiring such datasets is often impractical, costly and time-consuming. This bottleneck motivates the paradigm shift towards zero-shot anomaly detection [12], where models must detect and localize defects in previously unseen product categories without any specialized domain training data.

While existing benchmarks like MVTec AD [1] and VisA [25] have driven significant progress, they typically feature well-controlled, stable environments. Real-world industrial settings, however, confront models with severe visual interferences ranging from fluctuating illumination to complex material properties. The newly introduced MVTec AD 2 [9] dataset bridges this gap by bringing high-resolution imaging under diverse lighting conditions and challenging scenarios, thereby demanding highly robust, zero-shot capable visual inspection systems that can perform straight out of the box.

1.2. Challenge Description

The Zero-Shot Industrial Anomaly Detection track challenges participants to design robust algorithms capable of classifying and precisely localizing defects on entirely unseen products, without relying on any category specific training images.The primary objective is to achieve high anomaly segmentation and detection performance (evaluated via metrics such as SegF1 and AUROC) under significant real-world condition variations.

Participating in this zero-shot track involves overcoming several distinct, real world constraints:

1. Zero-Shot Generalization Without Priors: The complete absence of normal reference images for statistical calibration forces the model to inherently distinguish anomalies using generalized feature representations, commonly relying on the deep priors of foundation models (such as Vision Transformers [8] and Segment Anything Models [13]).

2. Robustness to Optical Complexities and Illumination: Real world environments are simulated through extreme

lighting variations and shifting of images. Transparent materials, highly reflective surfaces, and spatially random overlapping items heavily distort normal visual signatures, masking subtle defects.

2. Related Works

2.1. Foundation Models in Vision

The emergence of foundation models has significantly advanced various computer vision tasks, including zero-shot learning. Vision Transformers (ViT) [8] serve as the backbone for powerful models such as CLIP [16] and DINO [4]. CLIP focuses on aligning multimodal features, demonstrating exceptional semantic understanding. Meanwhile, DINO and its successors (e.g., DINOv3 [20]) produce high-quality, dense patch-level representations that are highly effective for dense prediction tasks. Similarly, the Segment Anything Model (SAM) [13] introduced prompt-based segmentation, showcasing unparalleled zero-shot generalization for boundary delineation across diverse domains.

2.2. Zero-Shot and Unsupervised Anomaly Detection

Traditional industrial anomaly detection heavily relies on one-class learning from normal reference images, exemplifed by methods like PatchCore [17], PaDiM [7], and AST [18]. To eliminate the dependency on clean training data, zero-shot and fully unsupervised approaches have gained attention. CLIP based methods, such as WinCLIP [12] and SAA [2], utilize text-vision alignment for zero-shot anomaly localization, while other approaches leverage few-shot adaptations [10, 22]. Recent advancements, instead of relying on manually curated normal sets, attempt to utilize unlabeled images directly. By modeling the manifold structure of high-dimensional image-level features [19], these methods refine coarse predictions, enabling fully unsupervised zero-shot anomaly classification and segmentation without human intervention.

2.3. Prompt-Driven and Cascaded Segmentation for Anomalies

While CLIP and DINO can effectively localize rough anomalous regions, they frequently struggle with precise boundary delineation due to resolution limits and interpolation artifacts. To address this, SAM has been integrated into anomaly detection frameworks [2, 21]. However, directly feeding naive bounding boxes or sparse points into SAM often leads to segmenting entire objects rather than the specific anomalous defects. To overcome these limitations, cascaded prompting strategies have been proposed. By leveraging coarse anomaly heatmaps (derived from foundation models) to sample highly probable positive and negative points, and iteratively refining the prompt inputs, SAM's extensive boundary-awareness can be fully exploited. Our integrated framework builds upon these insights, combining the precise regional scoring of mutually scored [14] unlabelled images with the robust boundary segmentation of a SAM-based cascaded refiner.

3. Methodology

3.1. Model Design

3.1.1. Approach

We propose a unified zero-shot industrial anomaly segmentation framework that combines multiscale statistical anomaly reasoning with prompt-guided spatial refinement. The method is designed specifically for the Industrial Track of the VAND 4.0 challenge, where models are required to generalize to unseen industrial anomalies without any training on the MVTec AD 2 dataset [9].

Our approach builds upon the mutual scoring philosophy of MuSc [14], where anomalous regions are identified through inconsistencies in patch-level feature similarity across images. While mutual-scoring methods are effective for localizing anomalous regions, the resulting anomaly maps are often spatially coarse and may contain noisy boundaries. To address this limitation, we integrate a cascaded SAM-based refinement stage that improves spatial consistency and boundary quality while preserving the original anomaly evidence.

Our proposed solution consists of three major stages:

1. First, each input image is passed through a frozen DINOv3 backbone [20] to extract multiscale patch-level features that capture both local texture and higher-level visual structure.
2. **Second,** the layer-wise DINOv3 patch tokens are passed to the LNAMD module from MuSc [14], which spatially aligns features from the selected transformer layers and aggregates them over multiple receptive-field scales to produce a normalized multiscale patch embedding for each image. These embeddings are then used as the input to the Mutual Scoring Module (MSM), which compares each patch embedding against patch embeddings from the other unlabeled images, converts weak cross-image similarity into patch-level anomaly scores, and outputs a coarse anomaly heatmap for each image.
3. **Finally,** the coarse anomaly heatmaps produced by mutual scoring are refined by a cascaded SAM module [13], which uses anomaly-aware prompts to generate sharper, boundary-aligned segmentation masks while preserving the original anomaly evidence.

The framework is fully class-agnostic and uses identical hyperparameters across all categories, complying with the challenge requirements. Furthermore, all backbone mod-

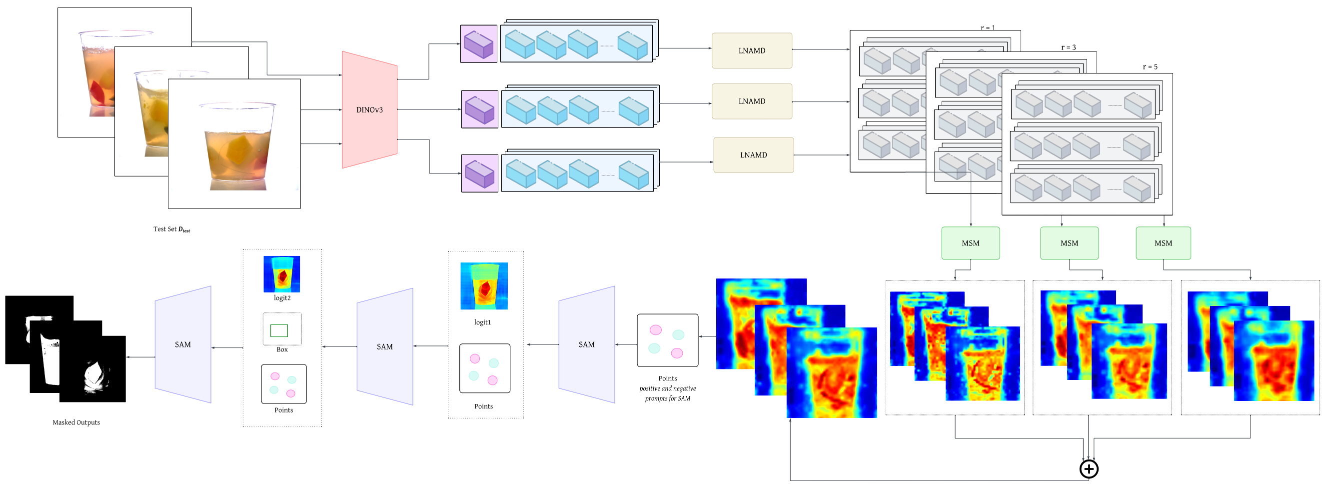

Figure 1. Overview of the proposed zero-shot industrial anomaly segmentation framework. Input images are first processed using a frozen DINOv3 backbone [20] to extract multiscale patch-level features from multiple transformer layers. The extracted features are passed through the LNAMD module and aggregated across multiple receptive-field scales following the MuSc framework [14]. The resulting embeddings are evaluated by the Mutual Scoring Module (MSM) to produce coarse anomaly heatmaps based on cross-image feature inconsistency. Finally, a cascaded SAM-based refinement stage [13] uses anomaly-aware prompts, including positive points, negative points, bounding boxes, and intermediate logits, to refine the coarse anomaly maps into spatially precise segmentation masks.

els remain frozen during inference, and no training or fine-tuning is performed on MVTec AD 2 [9].

3.1.2. Architecture

Feature Extraction. We replace the original CLIP backbone [16] used in MuSc [14] with the publicly available `facebook/dinov3-vitl16-pretrain-lvd1689m` model. DINOv3 [20] provides strong self-supervised dense visual representations that are well suited for industrial anomaly localization under varying lighting conditions and texture distributions.

Given an input image resized to 512×512, patch tokens are extracted from multiple transformer layers. Specifically, we use features from layers $\{6, 12, 18, 24\}$ to capture both low-level spatial structure and higher-level semantic information.

LNAMD Multiscale Embedding. The extracted patch tokens are processed using the LNAMD module. First, LNAMD strips away the non-spatial CLS token (the first token in the sequence) then takes the remaining flat sequence of N patch tokens and mathematically reshapes them into their original 2D spatial grid. Once the features are back in a 2D grid (Batch, Channels, Height, Width), following the original MuSc design [14], for each patch in a test image, LNAMD aggregates features from its local spatial neighborhood at multiple radii $r \in \{1, 3, 5\}$ by applying an average pooling operation to compress them into a single, neighborhood-aware feature descriptor.

All feature embeddings are L2-normalized before anomaly scoring.

Mutual Scoring Module (MSM). Following MuSc [14], the Mutual Scoring Module computes anomaly scores based on patch-level feature inconsistency across images. For each patch in an image, pairwise distances to patches from all other images are computed using Euclidean distance. Patches that exhibit weak similarity to neighboring image patches receive higher anomaly scores.

To improve scalability and reduce GPU memory consumption, we implement a chunked mutual-scoring strategy that computes pairwise distances incrementally instead of materializing the full pairwise distance matrix at once. This preserves the original mutual-scoring behavior while substantially reducing peak memory usage during inference.

The anomaly maps generated from different layers and receptive-field scales are averaged and then bilinearly upsampled to image resolution.

Cascaded SAM Refinement. While the mutual scoring mechanism (MuSc) [14] exhibits strong zero-shot anomaly localization capabilities, its resulting heatmaps are inherently coarse. Because the anomaly score is computed as a sum of patch-level distances interpolated to pixel space, the predictions often lack sharp, object-aware boundaries. To resolve this without introducing dataset-specific training, we propose a cascaded spatial refinement module leveraging the Segment Anything Model (SAM) [13].

The SAM module does not perform anomaly detection directly. Instead, it acts as a spatial refinement prior for the anomaly maps generated by the mutual-scoring stage.

Given a MuSc [14] anomaly heatmap, anomaly-aware prompts are automatically generated using:

- high-scoring positive points,
- low-scoring negative points sampled from a surrounding dilation ring,
- and a bounding box extracted from the highest-scoring connected component.

The refinement follows a three-pass cascade:

1. Point-based mask prediction.
2. Refinement using the previous SAM logits.
3. Final refinement using points, logits, and a bounding-box constraint.

The final SAM mask is used as a spatial gate over the original anomaly heatmap, preserving anomaly scores inside the refined region while suppressing responses outside the predicted anomaly area.

3.1.3. Prompting/Training

Our method operates in a strict zero-shot setting and does not use any images from the MVTec AD 2 dataset [9] during training.

All backbone models are used in their publicly available pretrained form:

- DINOv3 [20]: `facebook/dinov3-vitl16-pretrain-lvd1689m`
- SAM3TrackerModel [11]: `Sam3TrackerModel`

No fine-tuning, adapter training, prompt learning, or category-specific optimization is performed.

Unlike language-prompt-based VLM methods, our prompting strategy is entirely anomaly-aware and generated dynamically from the anomaly heatmaps during inference. Positive prompts correspond to highly anomalous regions, while negative prompts are sampled from surrounding low-response regions to suppress over-segmentation.

Preprocessing. All images are resized to 512×512 before inference. Standard ImageNet normalization is applied for DINOv3 feature extraction.

Thresholding Strategy. To determine the optimal binary segmentation threshold, we employ an F_1-maximization strategy calibrated on the MVTec AD [1] and VisA [25] datasets. Given the dense resolution of the predicted anomaly heatmaps, computing the exact precision-recall metrics over the entire dataset is computationally intractable and prone to memory exhaustion. To ensure stability, we extract a random, uniform subset of pixel-level anomaly scores and their corresponding ground-truth labels, subsampling 150,000 pixels per category. We evaluate this aggregated subset across the full continuous spectrum of predicted anomaly scores to construct a Precision-Recall (PR) curve. For each discrete threshold candidate τ, we compute the pixel-wise F_1 score:

$$F_1(\tau) = \frac{2 \cdot \text{Precision}(\tau) \cdot \text{Recall}(\tau)}{\text{Precision}(\tau) + \text{Recall}(\tau)}$$

The optimal threshold τ^* is selected as the argument that strictly maximizes this F_1 function. To adhere to the zero-shot, class-agnostic constraints of the evaluation framework, the pixel samples from all categories are concatenated to compute a single, dataset-wide global threshold. This unified global threshold is subsequently applied uniformly across all test categories to generate the final binary segmentation masks.

Evaluation Protocol. Raw anomaly heatmaps are used for pixel-level metrics such as AUROC and AUPRO to preserve comparability with the baseline MuSc framework [14]. SAM-refined masks are used only for SegF1 evaluation, where accurate binary segmentation boundaries are particularly important.

This separation ensures that the refinement module improves segmentation quality without altering the underlying anomaly scoring behavior of the mutual-scoring framework.

4. Results

4.1. Performance Evaluation

We evaluate the proposed method on the MVTec AD 2 benchmark [9] using a fixed threshold of 0.357525 for binary segmentation. Table 1 summarizes category-wise results across segmentation, pixel-level detection, image-level detection, precision-recall, and region-overlap metrics. The overall mean SegF1 is 27.78%, with mean pixel-level AUROC of 70.35%, image-level AUROC of 68.51%, pixel-level AP of 26.20%, image-level AP of 73.45%, AUPRO of 27.72%, and pixel-level F1 of 36.60%.

The results show that the method is most effective on categories with strong localized visual evidence. Fabric obtains the highest SegF1 score (55.55%), while walnuts achieves the strongest pixel-level AUROC (80.94%), AP-px (52.69%), and F1-px (60.78%). Vial records the highest image-level AUROC (94.67%), AP-im (98.55%), and AUPRO (58.20%), indicating strong image-level anomaly separability and region-level consistency. In contrast, can and wallplugs remain challenging, with low segmentation scores despite moderate pixel-level AUROC values, suggesting that coarse anomaly evidence does not always translate into accurate binary masks.

4.2. Comparison with Existing Methods

Table 2 compares different zero-shot anomaly detection approaches on MVTec AD 2 [9] using both pixel-level and

Table 1. Category-wise performance on MVTec AD 2 using threshold 0.357525. All values are reported in percent.

Category	SegF1	AUROC-px	AUROC-im	AP-px	AP-im	AUPRO	F1-px
Can	0.00	52.37	52.47	0.00	27.57	7.59	0.00
Fabric	55.55	87.96	48.37	43.94	51.47	22.45	57.20
Fruit Jelly	24.06	73.12	84.48	27.88	91.98	28.21	42.11
Rice	36.76	67.21	70.69	23.40	84.87	20.12	40.75
Sheet Metal	23.44	51.68	74.57	10.97	88.68	50.24	26.18
Vial	33.32	74.26	94.67	28.27	98.55	58.20	33.90
Wallplugs	6.70	75.26	42.57	22.49	55.86	11.14	31.86
Walnuts	42.41	80.94	80.24	52.69	88.58	23.80	60.78
Mean	27.78	70.35	68.51	26.20	73.45	27.72	36.60

image-level metrics, including SAA [2], WinCLIP [12], APRIL-GAN [5], AA-CLIP [24], AnomalyCLIP [23], AdaCLIP [3], VCP-CLIP [6], and SupAD [15]. Each entry is reported as a pair of scores. For pixel-level evaluation, the pair denotes (AUROC-px, AUPRO); for image-level evaluation, the pair denotes (AUROC-im, AP-im).

The comparison shows that MVTec AD 2 remains challenging for existing approaches. Among the listed baselines, VCP-CLIP achieves the strongest pixel-level AUROC (58.0%), while AnomalyCLIP obtains the highest baseline AUPRO (74.0%). At the image level, AdaCLIP reports the highest AUROC-im (85.4%), while APRIL-GAN achieves the strongest AP-im (47.7%). Our method improves the pixel-level AUROC to 70.35% and obtains a substantially higher image-level AP of 73.45%, indicating that the proposed feature representation and SAM-based refinement provide stronger anomaly evidence on MVTec AD 2.

Overall, the category-wise evaluation and the comparison against recent methods indicate that MVTec AD 2 remains a demanding benchmark for zero-shot anomaly localization. Strong image-level separability does not always guarantee high segmentation quality, making boundary-aware refinement and robust thresholding important directions for further improvement.

5. Discussion

5.1. Challenges Faced and Solutions

The zero-shot industrial anomaly detection setting presents several practical challenges. First, no category-specific normal images are available for learning product-dependent appearance statistics, which makes conventional one-class calibration unsuitable. To address this, our method relies on frozen foundation-model features and the mutual-scoring principle of MuSc [14], where anomalies are inferred from patch-level inconsistencies across the unlabeled image set rather than from a learned category template. This design allows the same inference pipeline and hyperparameters to be applied across all categories.

Second, MVTec AD 2 contains challenging acquisition conditions, including illumination changes, object shifts, reflective materials, transparent objects, and overlapping instances [9]. These variations can cause local feature responses to change even in normal regions, producing noisy anomaly maps. We mitigate this by extracting features from multiple DINOv3 layers [20] and aggregating them with LNAMD over several receptive-field radii. The resulting descriptors combine local texture information with broader contextual cues, making the anomaly score less dependent on a single layer or spatial scale.

Third, mutual scoring produces patch-level evidence that is useful for anomaly localization but often lacks precise object boundaries. This is especially problematic for SegF1, where small spatial errors can significantly reduce the score. We therefore use SAM-based refinement [13] as a boundary prior rather than as an anomaly detector. The coarse MuSc heatmap determines positive points, negative points, and bounding boxes, while SAM refines these prompts into sharper masks that preserve the original anomaly evidence but suppress spatially inconsistent responses.

5.2. Model Robustness

The proposed framework is designed to be robust in three ways. First, it is class-agnostic: all categories share the same backbone, scoring mechanism, and segmentation threshold, avoiding category-specific tuning that would vi-

Table 2. Comparison on MVTec AD 2. Pixel-level entries denote $(\mathrm{AUROC\text{-}px}, \mathrm{AUPRO})$ and image-level entries denote $(\mathrm{AUROC\text{-}im}, \mathrm{AP\text{-}im})$.

Metric	**SAA**	**WinCLIP**	**APRIL-GAN**	**AA-CLIP**	**AnomalyCLIP**	**AdaCLIP**	**VCP-CLIP**	**SupAD**	**Ours**
Pixel-level	(48.4, 66.5)	(55.5, 71.7)	(54.8, 70.6)	(56.7, 72.3)	(57.8, 74.0)	(55.6, 71.6)	(58.0, 72.1)	(56.2±0.3, 72.9±0.2)	**(70.35, 27.72)**
Image-level	(63.7, 13.6)	(70.0, 31.3)	(79.8, 47.7)	(80.6, 45.3)	(72.3, 37.3)	(85.4, 30.5)	(80.8, 45.8)	(81.2±0.2, 46.1±0.3)	**(68.51, 73.45)**

olate the zero-shot setting. Second, the use of frozen DINOv3 and SAM3TrackerModel [11] avoids training instability and reduces the risk of overfitting to the public validation categories. Third, the chunked implementation of MSM improves computational robustness by reducing peak memory usage while preserving the original pairwise scoring behavior.

The results indicate that this design provides strong pixel-level anomaly evidence, particularly for categories with localized defects and discriminative texture deviations. However, robustness is not uniform across all categories. Objects with weak contrast, highly reflective surfaces, or ambiguous normal variations remain difficult because feature inconsistency may not align perfectly with the true defect region. In these cases, SAM refinement can improve boundary quality only when the initial heatmap provides reliable prompts; if the coarse heatmap is diffuse or misplaced, refinement may preserve or amplify the initial localization error.

5.3. Future Work

Several directions could further improve the method. First, prompt generation for SAM could be made more adaptive by considering heatmap uncertainty, connected-component stability, and multi-threshold agreement rather than relying on a fixed set of high- and low-response regions. Second, the thresholding strategy could be improved with calibration methods that remain class-agnostic but better account for score-distribution shifts across material types and lighting conditions. Third, future work could explore more efficient approximate nearest-neighbor or prototype-based variants of mutual scoring to reduce inference time on large image sets without sacrificing localization quality.

Finally, the current framework treats feature extraction, mutual scoring, and mask refinement as separate stages. A promising extension is to introduce feedback between the refinement mask and the anomaly-scoring stage, allowing the system to re-score uncertain regions after spatial refinement. Such an iterative design may improve performance on small, low-contrast, or partially occluded defects while preserving the zero-shot nature of the approach.

6. Conclusion

We presented a zero-shot industrial anomaly segmentation framework that combines MuSc-style mutual scoring with dense DINOv3 representations and cascaded SAM-based spatial refinement. The method is designed for settings where category-specific normal data and defect annotations are unavailable, making it suitable for real-world inspection scenarios with changing products, limited calibration data, and diverse imaging conditions.

Our results on MVTec AD 2 show that the proposed pipeline produces meaningful anomaly localization under challenging acquisition conditions. In particular, the method achieves a mean SegF1 of 27.78%, while obtaining strong category-level results such as 55.55% SegF1 on fabric, 80.94% pixel-level AUROC on walnuts, and 94.67% image-level AUROC on vial. These findings indicate that combining foundation-model features with prompt-driven mask refinement can improve the conversion of coarse anomaly evidence into spatially useful segmentation outputs.

Beyond the reported scores, the main significance of the framework lies in its training-free and class-agnostic design. By avoiding category-specific fine-tuning, the same inference pipeline can be applied across unseen industrial objects, which is important for practical deployment in manufacturing environments. The results also suggest that vision foundation models and segmentation models can serve complementary roles in anomaly detection: dense visual representations provide robust anomaly cues, while promptable segmentation models improve boundary quality and region consistency.

Future work should focus on improving threshold robustness, reducing failure cases on subtle or low-contrast defects, and developing stronger prompt generation strategies for complex object geometries. Extending this direction may further advance zero-shot and few-shot anomaly detection systems that rely on vision-language models and foundation segmentation models for real-world industrial inspection.

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